# SAATHI: a chair that warns of worsening heart failure

**-Team Wildcat**

## Description

SAATHI is an ordinary armchair with a daily heart check built in: one button, 90 seconds sitting still, three spoken yes/no questions asked during the measurement (110–116 s by design, not yet timed with users). On a **locked set of 200 never-seen synthetic patients, scored once**, it caught **80% of 129 worsening events** (95% CI 72–86%), left **95.8%** of normal days alone, gave **0.77 false alerts per 30 normal patient-days** (0.49 per 30 recorded patient-days) and a median **3 days** of warning for caught events; **40%** of alert episodes were real worsening. Parts cost **₹4,661** and software, firmware and both circuits are built and tested in simulation.

## Method

| Signal | Sensor and position | Weight |
|---|---|---|
| Body weight | 4 load cells (one bridge) in the legs + HX711; the footrest hangs on the frame, so the feet are weighed | 1.0 |
| Foot bioimpedance (exploratory) | 50 kHz 4-electrode circuit, 4 stainless plates on the fold-out footrest | 1.0 |
| Resting heart rate | Seat PVDF film (BCG) and right-armrest MAX30102 (PPG), fused | 0.7 |
| Breathing rate; $SpO_2$ (trend) | MAX30102 PPG | 0.7; 0.5 |
| 3 symptom questions | Speaker; green/red buttons on the left armrest | 0.7 per yes |
| Pulse transit time; chair use | BCG-to-PPG timing; sit-downs from the load cells | 0.25 each |

**Hardware.** ESP32, offline storage, SIM800L SMS from a flash outbox (alerts never evicted, partial messages resume, retried until the network confirms; delivery is not guaranteed without network), 2 × 18650 cells (can provide about 20 days of power without charging). Two custom circuits simulated in LTSpice: the seat BCG front-end ( For penetrating through clothing and secondary verification) and the foot-impedance circuit (Foot current ≤ 1.65 µA DC and ≤ 51 µA rms, This makes it always safe. ).

**Signal processing with quality gates.** Weight is read from the highest steady plateau (We reject it if the person was not at full load for half the check, if foot contact failed, or out of range); heart rate from BCG and PPG peaks; $SpO_2$ from per-beat red/IR ratios; breathing rate from three PPG modulations; PTT. Finger presence, beat consistency and estimate agreement are checked, so a broken or badly used sensor gives “no reading”, never a number. The ESP32 firmware runs the same code (dsp.cpp, detector.cpp).

**Personal baseline and alert.** The first 14 days set each patient’s normal (median and spread); it keeps updating from calm days at least 11 days old, so a new diuretic dose becomes the new normal but a 3–10 day worsening does not. Points increase only beyond one spread, in the worsening direction. Alert: a high score on 2 measured days in a row, or weight +2 kg in 3 days, or $SpO_2$ 4+ points below the person’s own normal twice, or checks stopping for 2 days after a high score. A weight > 8 kg off never suppresses the day: the chair asks who sat and, without an answer, scores everything except weight. Every day has a status; missing data is never shown as normal; reasons are observational (“foot impedance −4% vs usual”).

## Evidence

Studies show no single signal is enough (we rely on their abstracts). Daily weight [1]: gain precedes many admissions, not all. Chest impedance from implants [2] falls before admission; our foot version is a corollary of the same research paper, made to work more comfortably. Wearable seismocardiography [3] (a chest patch sensing heart vibration) tracked heart-failure status; our seat BCG is a related measurement, used only for heart rate. Heart rate is also verified using a PPG sensor. Multi-signal personal baseline [4]: 76–88% caught, about 6.5 days early, with a chest patch: our benchmark.

## Validation and results

**Data we generated:** Synthetic patients with a personal normal per signal; worsening starts 3–10 days before admission. Types are: congestive, no weight gain and subtle look-alikes are built in (festival meals, fevers, diuretic changes, heavy clothing, missed checks, poor contact, pacemakers) to simulate real life variation. Every check is simulated as raw 90-second waveforms and processed by the pipeline.

**Protocol.** Development: 120 patients (5,951 patient-days), split by patient 59/61; weights were fixed in advance; only the threshold (1.25, lowest keeping training false alerts ≤ 1 per 30 normal patient-days) and the time-of-day correction were learned on training patients. The development test set was scored by three detector versions after two reviews (74% / 95.8% / 0.83 for the final one), so the headline comes from a locked set: protocol, seed and file hashes recorded before 200 new patients (10,151 patient-days) were generated; scored once; a repeat is refused.

| Locked test | Alert | No alert |
|---|---|---|
| Event (per event) | TP 103 | FN 26 |
| Normal day (per day) | FP 276 | TN 6,249 |

| Measure | Result |
|---|---|
| Event (per event) | Sensitivity 80% (72–86%) |
| Normal day (per day) | Specificity 95.8% |
| Alert episodes | 113 real, 167 false |
| Precision; false per 30 normal days* | 40%; 0.77 |
| Warned ≥ 1 / 3 / 5 / 7 days (of all 129 events) | 80% / 50% / 19% / 8% |

*Normal patient-days: after the 14-day learning period, outside any worsening window; per 30 recorded patient-days the same episodes give 0.49 (a different metric). The lock is a reproducibility safeguard (file hashes, a logged refusal of a second run), not an independent audit.

By type: congestive 97%, no weight gain 72%, subtle 36%. Robustness: 10 new datasets 77% (71–84%); stress tests on patterns not designed for: slow 14–30-day worsening 96%, late surge 65%, patients who skip checks when unwell 65%, everything hard at once 53%. In Patients with Pacemakers and artificial devices, The Foot Impedance signals could be life threatening, so we disable them, hence the Lite option. Faults in raw waveforms (14 kinds): before the quality gates, a missing finger gave a wrong $SpO_2$ in 95% of checks and leaning on the floor a wrong weight in 100%; after, 0% wrong for every fault except a partly failed load cell (caught downstream only above 8 kg). Firmware = Python on every patient-day and on the same raw waveforms. Real recordings (public PhysioNet [5,7,8]): beat detector within 0.03 bpm of expert ECG labels; pulse-wave heart rate within ±5 bpm in 98% of usable windows; PPG breathing rate weak (few usable windows).

## Limits

**Not clinical evidence.** We wrote the data and the detector; real patients will differ. No physical chair: weight accuracy, the real seat signal, battery life and usability are unmeasured (hardware/BENCH_TESTS.md).

**Many false alarms and short warning.** Precision 40%; half of events get 3+ days; subtle worsening mostly missed.

**Weak signals.** PPG breathing rate; $SpO_2$ is a trend; foot impedance unproven. BIDMC run and native-speaker prompt review pending. Budget margin ₹88 (conservative).

**Disclaimer.** SAATHI is a monitoring aid, not a medical diagnosis tool or a replacement for clinical advice. All data are computer-generated. SOS texts family; it is not an emergency service. Code, tests and setup**:**
**https://github.com/Shan-boop-cmd/Saathi_Chair**

**References.** [1] Chaudhry SI et al. Circulation 2007;116:1549–54, doi:10.1161/CIRCULATIONAHA.107.690768. [2] Yu CM et al. Circulation 2005;112:841–8, doi:10.1161/CIRCULATIONAHA.104.492207. [3] Inan OT et al. Circ Heart Fail 2018;11:e004313, doi:10.1161/CIRCHEARTFAILURE.117.004313. [4] Stehlik J et al. Circ Heart Fail 2020;13:e006513, doi:10.1161/CIRCHEARTFAILURE.119.006513. [5] Goldberger AL et al. Circulation 2000;101:e215–20. [6] Baim DS et al. J Am Coll Cardiol 1986;7:661–70 (BIDMC CHF). [7] Moody GB, Mark RG. IEEE EMB Mag 2001;20:45–50. [8] Clifford GD et al. Comput Cardiol 2015;42:273–6. Full list: REFERENCES.md. in GitHub repo.